

Does energy saved by powering data centers with AI offset energy cost of AI?

Posted by u/Lippopa • 0 points • 14 comments

I am doing an assignment on AI data center's sustainability and cant figure out from research if the total energy saved by AI making data centers more efficient is enough to cover the energy lost by powering the AI itself?

Comments

u/Mercury-68 • 1 points • 2025-01-09 03:08 UTC

No. It is not the objective of AI energy saving.

u/Remarkable-Coffee535 • 1 points • 2025-01-09 01:41 UTC

It's an interesting article but not anything I've heard about before. I think it's probably pretty speculative so that's why it's hard to find data. In the end, though, since AI consume so much more power than cloud, any efficiencies it creates would probably be negated by the increased use of AI in general so I don't see the power demands going down at all.

u/geekworking • 1 points • 2025-01-08 22:52 UTC

It depends on what you consider "AI".

If you look at AI as the meaningless marketing term sense where it's slapped on every existing product to make it "new," then "powering the datacenter will AI" just means that they have some sort of building control software. The software vendor added a few fancy reports, and it maybe needs to resources of a single gaming PC. This could offset itself.

If you consider "AI" as all of the clusters of thousands and thousands of super power-hungry GPU servers that run something like ChatGPT, then not even close.

u/Evil_Lord_Cheese • 3 points • 2025-01-08 22:45 UTC

Simple- No.

We are already very good at efficient designs, and the benefits of AI optimisation of the cooling systems is ~10-15% improvement (I think Google published some results). That does not mean the whole site is 10-15% better as the cooling system is typically only 15-25% of the site, so realistically overall savings are <5% from the AI there.

Compare that to the fact companies are building entire new data centre campuses purely to support

AI/ML workloads shows the balance between the two factors is so lopsided to get even close to parity every single cloud DC out there would need to be AI optimised, and that is faaaaar from the case.

u/tokensRus • 1 points • 2025-01-08 22:36 UTC

To calculate the environmental impact of a datacenter ai or not, is a highly structured, standardized and complex process. Maybe you should read some blogs or whitepapers about principles of ESG for datacenters or for carbon footprint reporting. Check out the blog of Schneider Electric if you like, they provide good whitepapers about dc efficiency and environmental impact

u/Lippopa • 1 points • 2025-01-08 23:11 UTC

Thank you!

u/longwaybroadband • 6 points • 2025-01-08 22:16 UTC

AI not only consumes a shit load of energy but water as well...every time you search one time that's a glass of water used to cool the dc's

u/Corbusi • 1 points • 2025-01-08 22:12 UTC

AI begets more processing power begets more electricity begets more heat waste begets more earth death

u/ElevenNotes • 8 points • 2025-01-08 21:45 UTC

No idea how and why you think AI saves energy?

u/Lippopa • 1 points • 2025-01-08 22:13 UTC

I am aware and the recreational use of AI is mentioned negatively in the use section of the life cycle in my assignment alongside it increasing general data center efficiency by intelligently allocating resources. There's plenty sources for overall carbon footprint of AI but I'm struggling to find exact statistics on carbon footprint improvement when AI is used for this efficiently at places like google, as I'm figuring out how much this could potentially offset the environmental damages of AI itself. <https://deepmind.google/discover/blog/deepmind-ai-reduces-google-data-centre-cooling-bill-by-40/>

u/Remarkable-Coffee535 • 2 points • 2025-01-09 01:41 UTC

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u/VSOLPro-James • 1 points • 2025-01-09 21:13 UTC

Unless AI gets us to clean, low cost energy.

u/Candid_Ad5642 • 1 points • 2025-01-09 05:20 UTC

>AI consume so much more power than cloud

Apples and orangutans

But anyways, wouldn't an AI workload consume about the same amount of power whether it's on prem, or in a public cloud? You just won't see it in a public cloud

There is no cloud, it's just someone else's computer

u/Remarkable-Coffee535 • 2 points • 2025-01-11 12:57 UTC

Let's zoom out, I'm saying a datacenter running mostly AI servers will consume a lot more power than one running cloud that's the same size. AI could help us discover greater efficiencies but the act of migrating to AI applications as a whole will consume so much power that those efficiencies won't matter. It's like saying cars from 80 years ago were 3x less efficient in gas mileage but now there are a lot more cars on the road so overall gas consumption is much higher today even those we've gotten more efficient

u/Lippopa • 1 points • 2025-01-08 21:48 UTC

[<https://www.digitalrealty.co.uk/resources/articles/sustainable-data-centre-ai>]
(<https://www.digitalrealty.co.uk/resources/articles/sustainable-data-centre-ai>)

was reading into this

u/Ambitious_Budget_671 • 13 points • 2025-01-08 21:43 UTC

No.